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Technical Specification

Digital cellular telecommunications system (Phase 2+); Explicit Call Transfer (ECT) supplementary service; Stage 2 (GSM 03.91 version 6.0.0 Release 1997)





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Foreword

This Technical Specification (TS) has been produced by the Special Mobile Group (SMG).

The present document defines stage 2 of Explicit Call Transfer (ECT) supplementary services within the digital cellular telecommunications system (Phase 2/Phase 2+).

The contents of the present document is subject to continuing work within SMG and may change following formal SMG approval. Should SMG modify the contents of the present document it will be re-released with an identifying change of release date and an increase in version number as follows:

Version 6.x.y

where:

- 6 indicates Release 1997 of GSM Phase 2+
- x the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- y the third digit is incremented when editorial only changes have been incorporated in the specification.

1 Scope

The present document gives the stage 2 description of the call transfer supplementary services.

Only one call transfer supplementary service has been defined, this is the Explicit Call Transfer (ECT) supplementary service, and is described in the present document.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.
- A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.
- For this Release 1997 document, references to GSM documents are for Release 1997 versions (version 6.x.y).

[1] GSM 01.04: "Digital cellular telecommunications system (Phase 2); Abbreviations and acronyms".

[2] GSM 03.83: "Digital cellular telecommunications system (Phase 2); Call Waiting (CW) and Call Hold (HOLD) supplementary services - Stage 2".

[3] GSM 04.08: "Digital cellular telecommunications system (Phase 2); Mobile radio interface layer 3 specification".

[4] T/S 22-21 (version 9): "Integrated Services Digital network (ISDN); Explicit Call Transfer (ECT) supplementary service; Functional capabilities and information flows".

[5] DE/SPS-6001.22 (version 9): "Integrated Services Digital network (ISDN); Explicit Call Transfer (ECT) supplementary service; Signalling System No. 7; Integrated services digital network User Part (ISUP) protocol".

3 Definitions and abbreviations

3.1 Definitions

First call: One of the subscriber A calls (answered).

Notification Indicator (NI): Indicates to each remote party in which state of the other remote party ECT was performed (active, alerting).

Redirection Number (Rdn): Includes the presentation indicator and the directory number of the other remote party.

Second call: The other subscriber A call (answered or alerting).

Subscriber A (PARTY A): The served mobile subscriber - the one who has subscribed to, and invokes the ECT Supplementary Service.

Subscriber B (PARTY B): The other party in the subscriber A first call.

Subscriber C (PARTY C): The other party in the subscriber A second call.

Subscriber D (PARTY D): The forwarded-to party when the call is forwarded by the subscriber C.

Transferred call: The resulting call after successful explicit call transfer between B and C.

3.2 Abbreviations

In addition to those below, abbreviations used in the present document are listed in GSM 01.04.

ECT:	Explicit Call Transfer supplementary service
LI:	Line Identity
NI:	Notification Indicator
Rdn:	Redirection number
RdnB:	Redirection number of the party B
RdnD:	Redirection number of the party D

4 Explicit Call Transfer (ECT)

4.1 Functions

The following function has been identified for the explicit call transfer service:

MAF027

Explicit Call Transfer related authorizations examination

The ability of a PLMN component to determine the authorizations relating to explicit call transfer. See figure 1.

Location: VLR

Within the authorization examinations diagram, the messages shown to and from the left are to and from the MSC.

After receiving a "invoke ECT request" the VLR will check if the Explicit Call Transfer service is provisioned for the served subscriber. If the service is provisioned the VLR send back to the MSC "Explicit Call Transfer available". If the service is not provisioned the VLR will send back to the MSC "Explicit Call Transfer not available". When the response has been sent back to the MSC the process will return to the idle state.

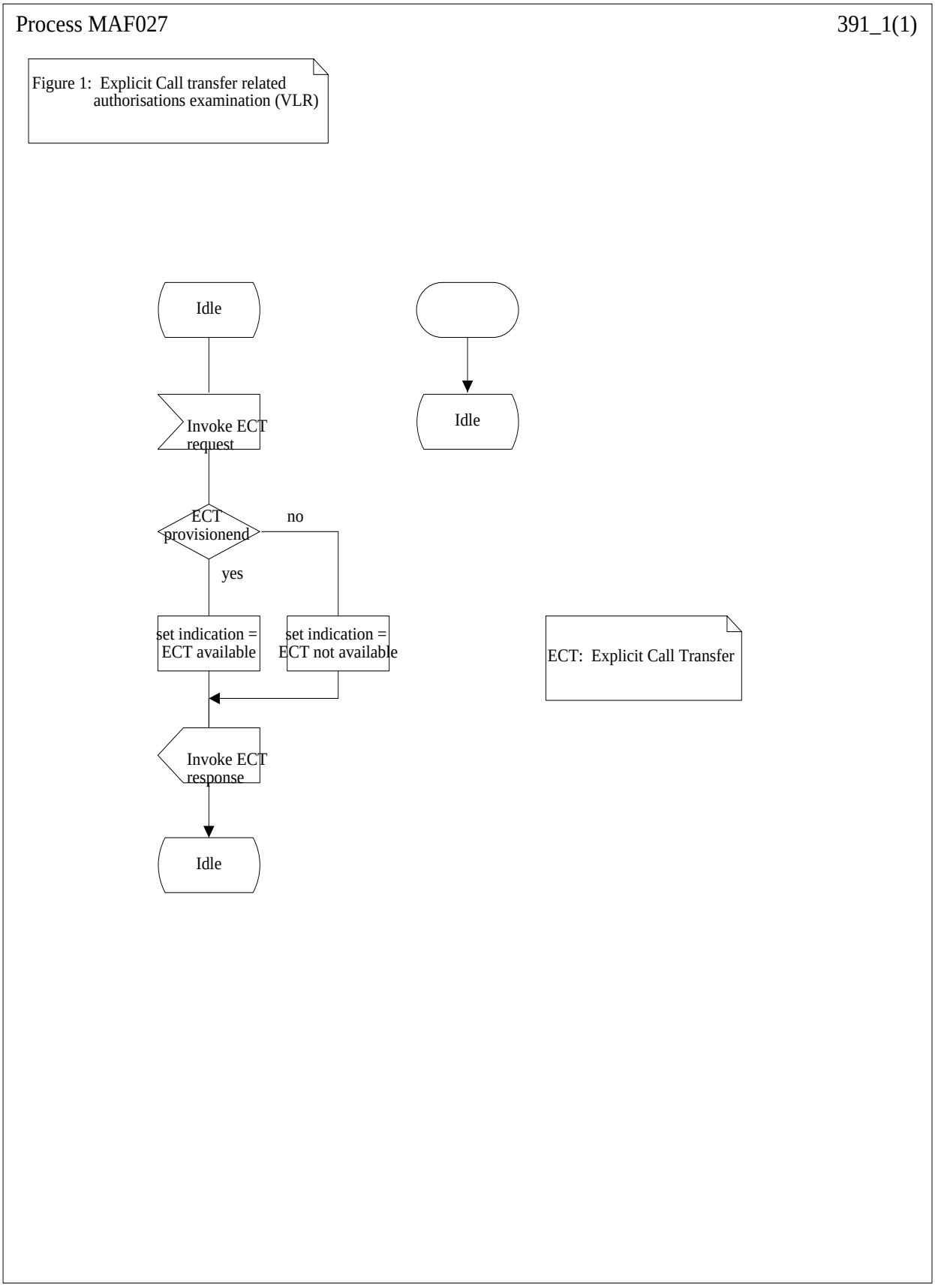


Figure 1: Explicit Call Transfer related authorizations examination (VLR)

4.2 Overall SDL-diagrams and information flows

4.2.1 General description

The overall SDL-diagrams represent the network as a whole. The overall SDL-diagrams show the status of the service as perceived by the served mobile subscriber, as well as the status as perceived by any of the other parties. Besides this, the overall SDL-diagrams show the actions to be taken by the network and the information provided by the network to the users.

Within the overall SDL-diagrams, messages to and from the served mobile subscriber are indicated to and from the left, whereas messages to and from remote parties are indicated to and from the right.

The following states for the invocation of the ECT supplementary service are defined:

a) First Call (Active, Held), Second Call (Active, Idle);

b) Second Call (Active, Held), First Call (Active, Idle);

c) First Call (Active, Held), Second Call (Call Delivered, Idle).

These two dimensional states are also used in the SDLs and information flows:

- the first dimension is a normal basic call state "active" or "call delivered";
- the second dimension is "held" meaning that the call is set on hold.

NOTE: The call state "call delivered" means that a ALERT message has been received in the MS, but no ANSWER MESSAGE (ANM) is received.

In the information, flows it is assumed that the served subscriber is a mobile subscriber and that the other parties are mobile or fixed subscribers.

Party A is the subscriber controlling the Explicit Call Transfer Call (served mobile subscriber). Party B is the first remote party called. Party C is the second remote party called.

The served party is disconnected by the generic disconnect/release procedure after a successful transfer request. The connection of the remote party calls in a new call (transferred call) is located in the MSC.

The first figures of the information flows (figures 4 and 7) show the unsuccessful case of the transfer request (check in the VLR or in the MSC fails).

The second figures (figures 5 and 8) show the successful case of the transfer request.

4.2.2 ECT (both calls answered)

After receipt of a ECT request from the served subscriber, the MSC/VLR will check if the Explicit Call Transfer supplementary service is provisioned for the served subscriber (see also MAF027).

If the ECT Supplementary Service is provisioned, then the MSC/VLR will check if the transfer is barred by virtue of call states or supplementary service interactions (see also figure 3 and subclause 4.3).

If there are no such barring causes then the MSC/VLR also checks if CUG restrictions are infringed (see also figure 3 and subclause 4.3).

If the outcome of these checks are successful and the loop prevention option is supported, an endless loop prevention check shall be performed according to T/S 22-21 (version 9) and DE/SPS6001.22 (version 9).

If the result of this check is also successful the both calls shall be connected in the MSC (without including the served subscriber in this connection), the held party will be retrieved and both remote parties will be notified that call transfer was done.

With this notification the both subscribers will be informed about the state of the other remote party in which call transfer was done ("call transferred, active") and if possible about the identity (Redirection number) of each other (for details see also subclause 4.3).

After that the served mobile subscriber will be disconnected from both calls.

If the checks fail the ECT request will be rejected and the two calls remain in the call states in which they were before ECT was attempted.

The overall SDL for Explicit Call Transfer (both calls are answered) is shown in figure 2.

The checks if Explicit Call Transfer is barred or not are shown in figure 3.

The corresponding information flows are given in figure 4 and figure 5.

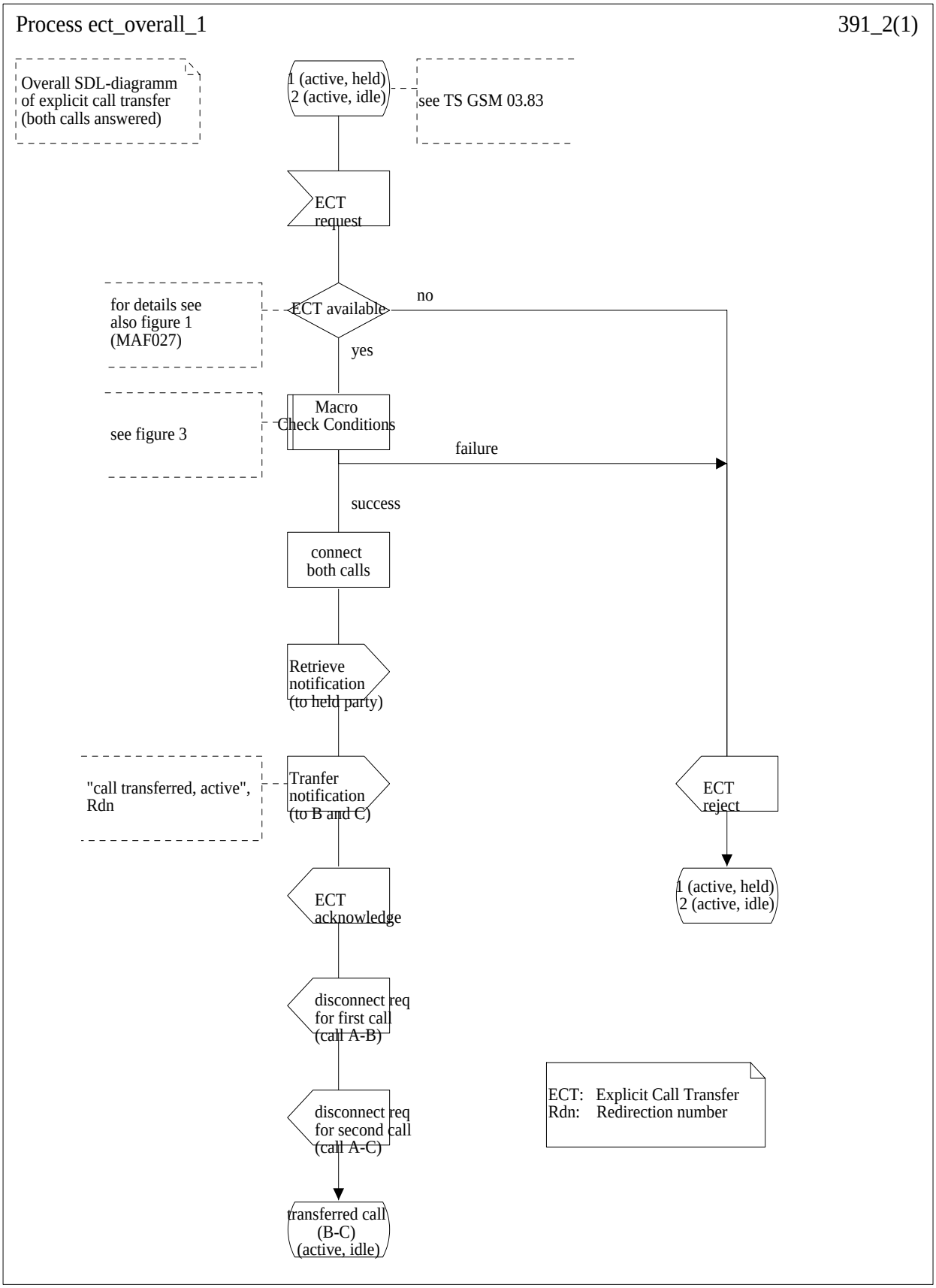


Figure 2: Overall SDL-diagram of Explicit Call Transfer (both calls answered)

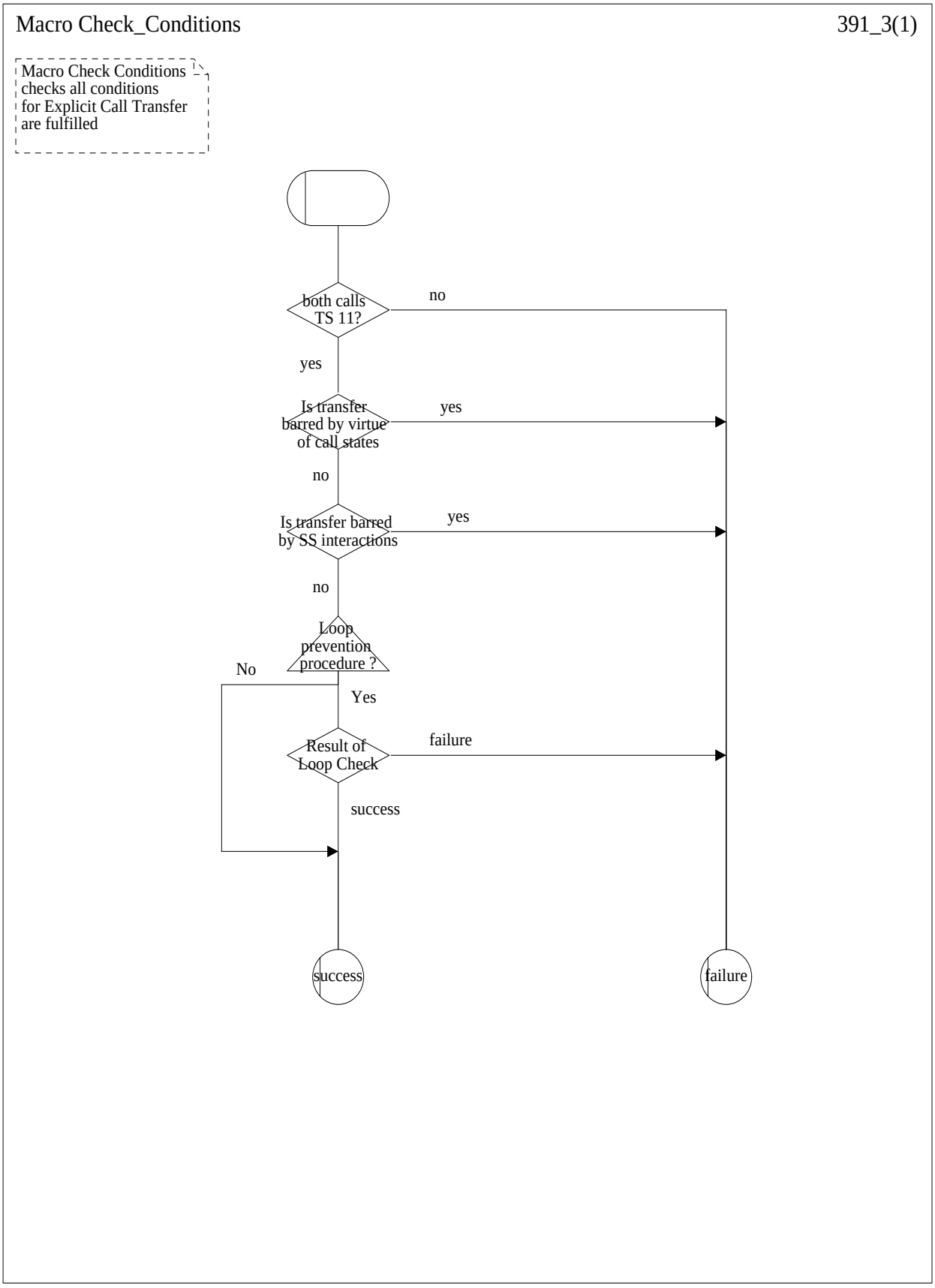
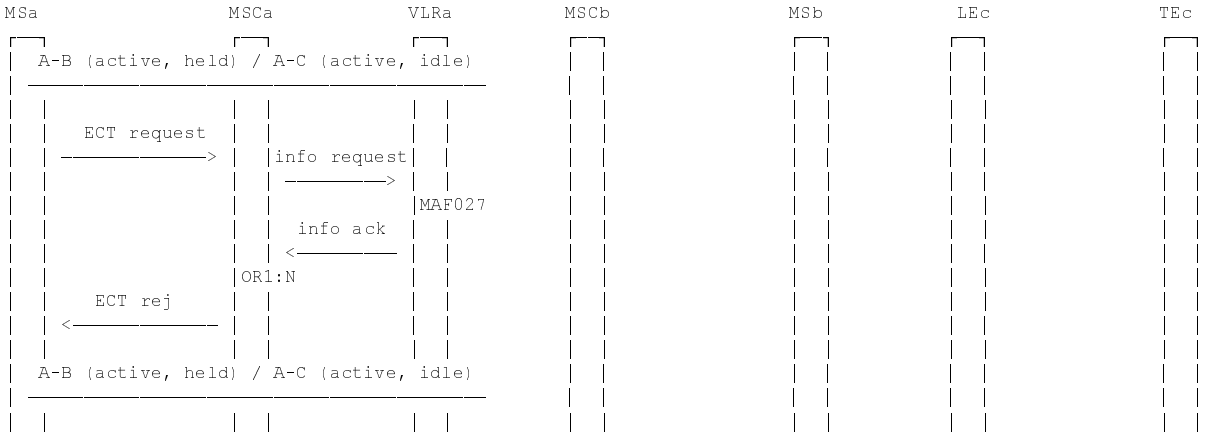
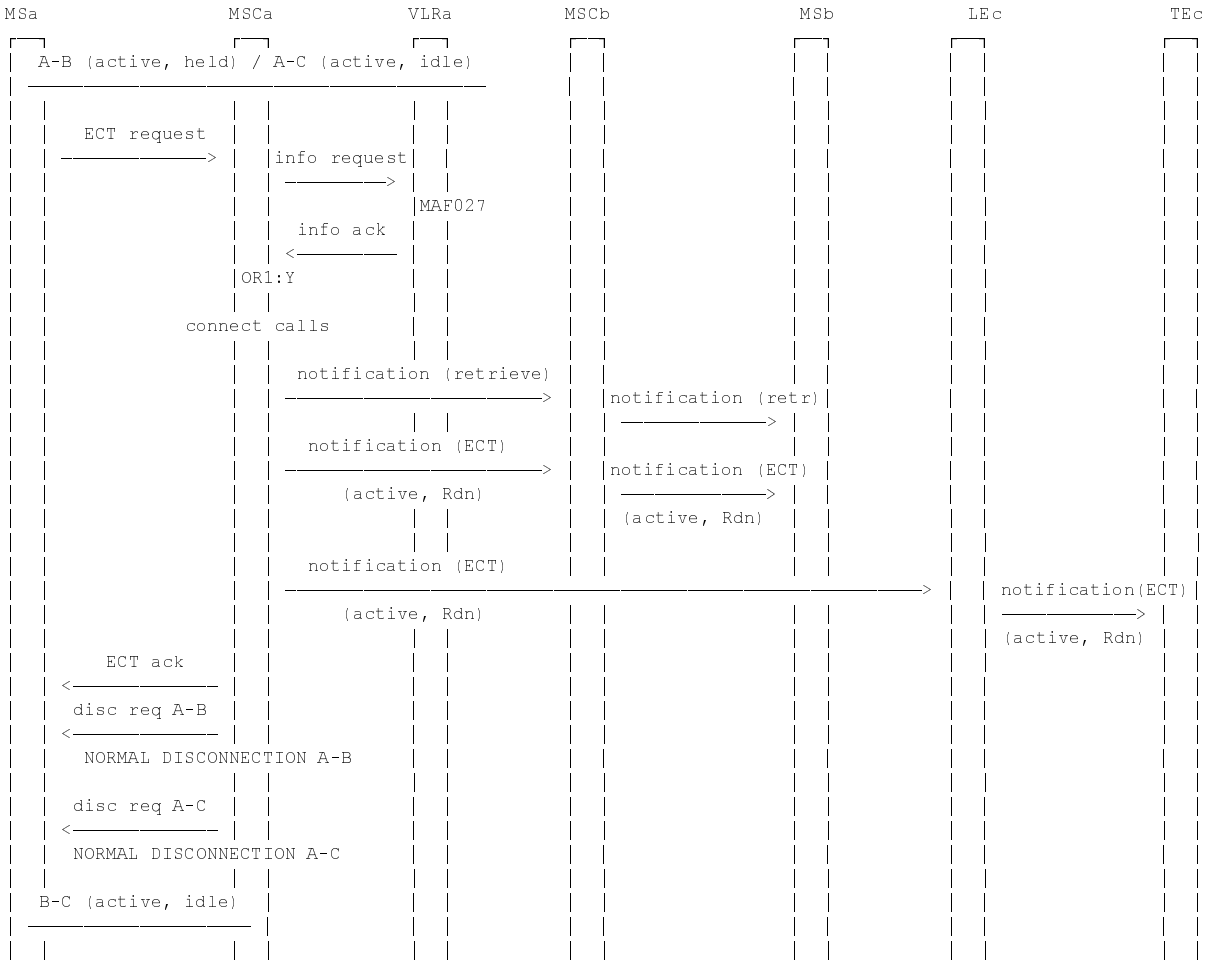


Figure 3: Macro Check Conditions



NOTE: OR1: Checks in VLR and MSC ok? (Y: yes N: no).

Figure 4: Information flow for failed explicit call transfer request (both calls answered)



NOTE: OR1: Checks in VLR and MSC ok? (Y: yes N: no).

Figure 5: Information flow for successful explicit call transfer (both calls answered)

4.2.3 ECT (one call answered, the other alerting)

In this case, generally the same procedures will apply as in the other case (both calls answered). The same checks shall be performed and if all checks are fulfilled both calls shall be connected together (without including the served subscriber in this connection). After the connection of the both calls, both subscriber (B and C) will be notified about the call transfer invocation in the same way as in the case where the two calls are answered.

The transfer notification to the subscriber B will include the information that the transfer was done in the altering state of subscriber C ("call transferred, alerting"). After receipt of the answer message from subscriber C, the subscriber B will be notified again, indicating that answer has taken place subsequent to the alerting transfer ("call transferred, active").

The overall SDL for Explicit Call Transfer (one call answered, the other alerting) is shown in figure 6.

The corresponding information flows are given in figure 7 and figure 8.

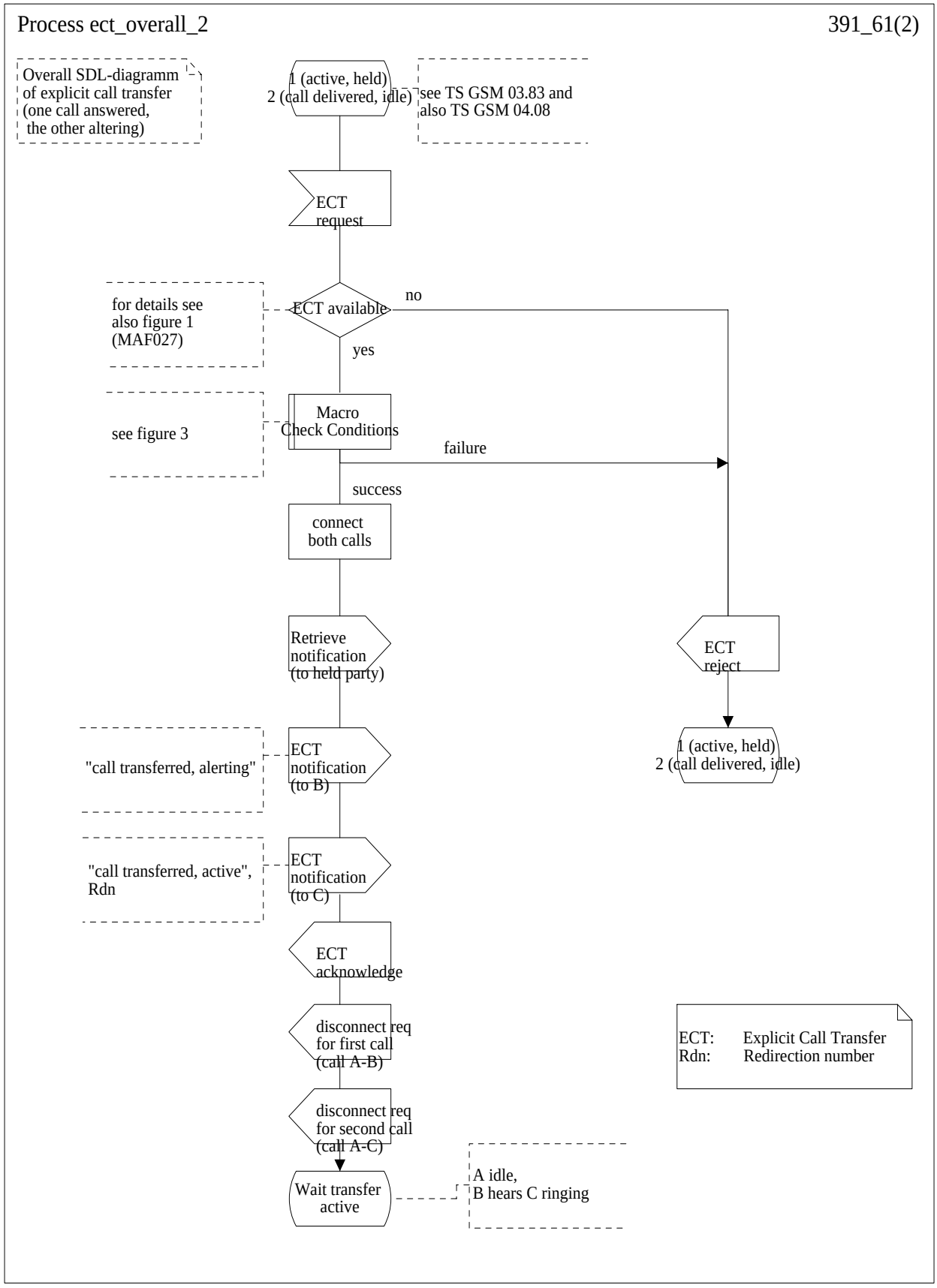


Figure 6: Overall SDL-diagram of explicit call transfer (one call answered, the other alerting) (page 1 of 2)

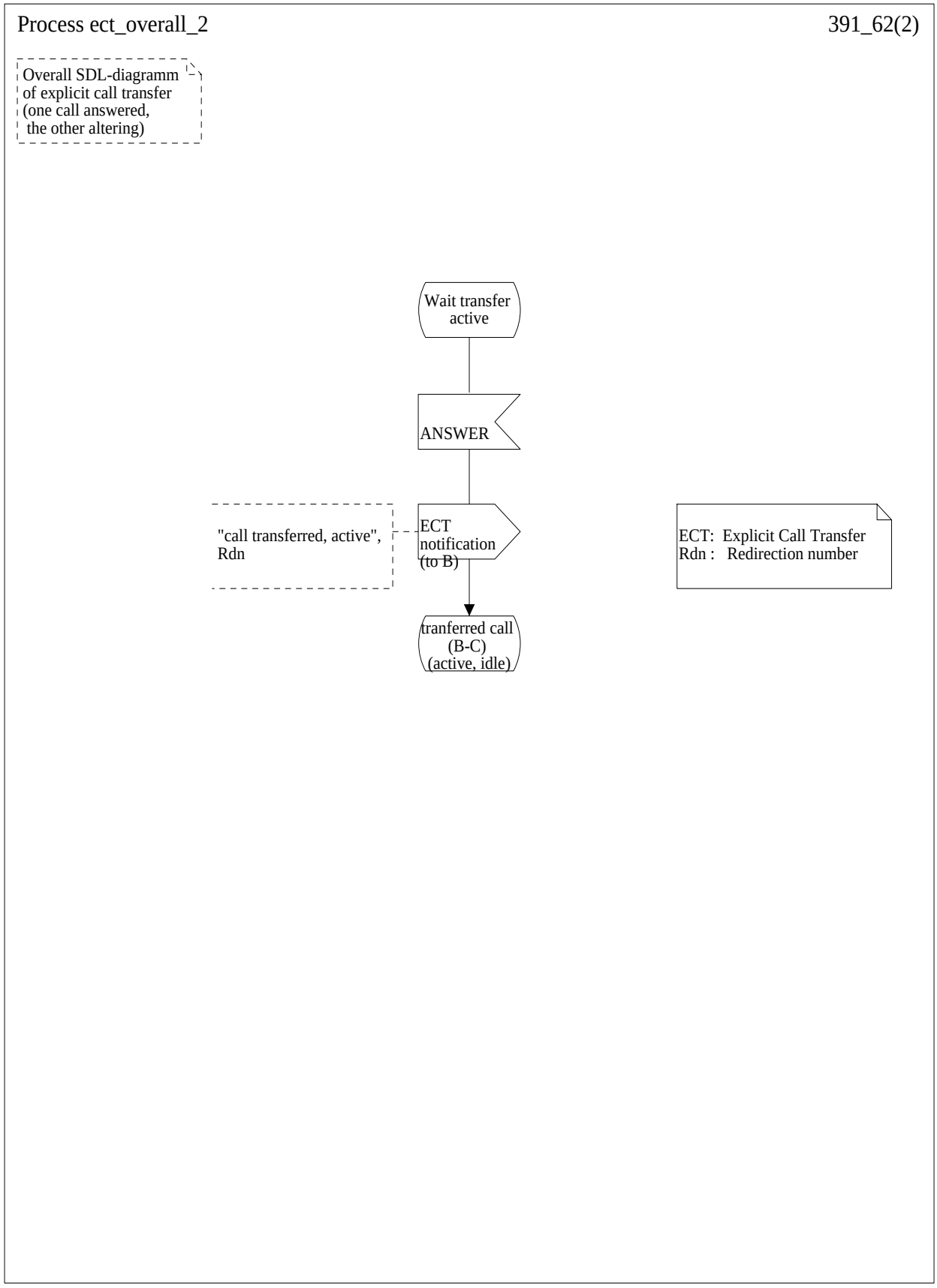
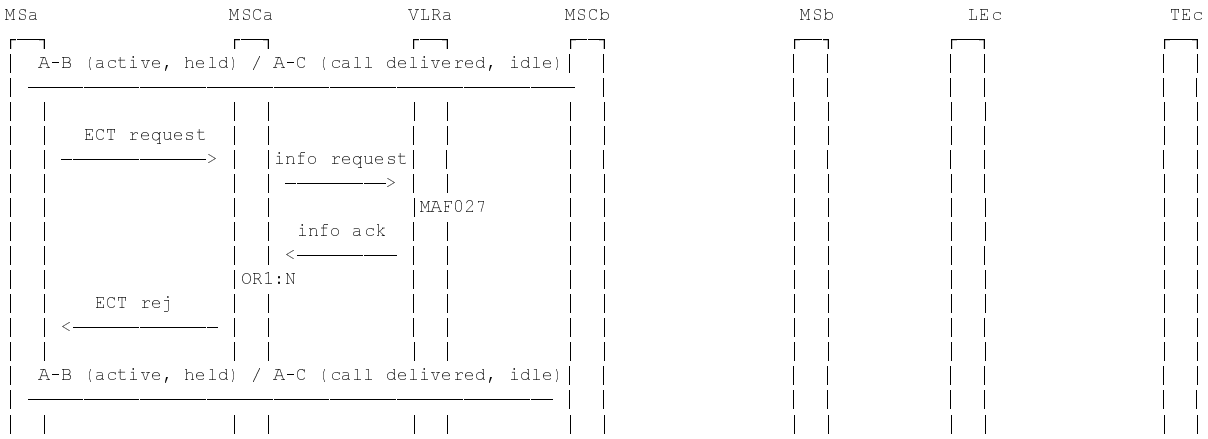
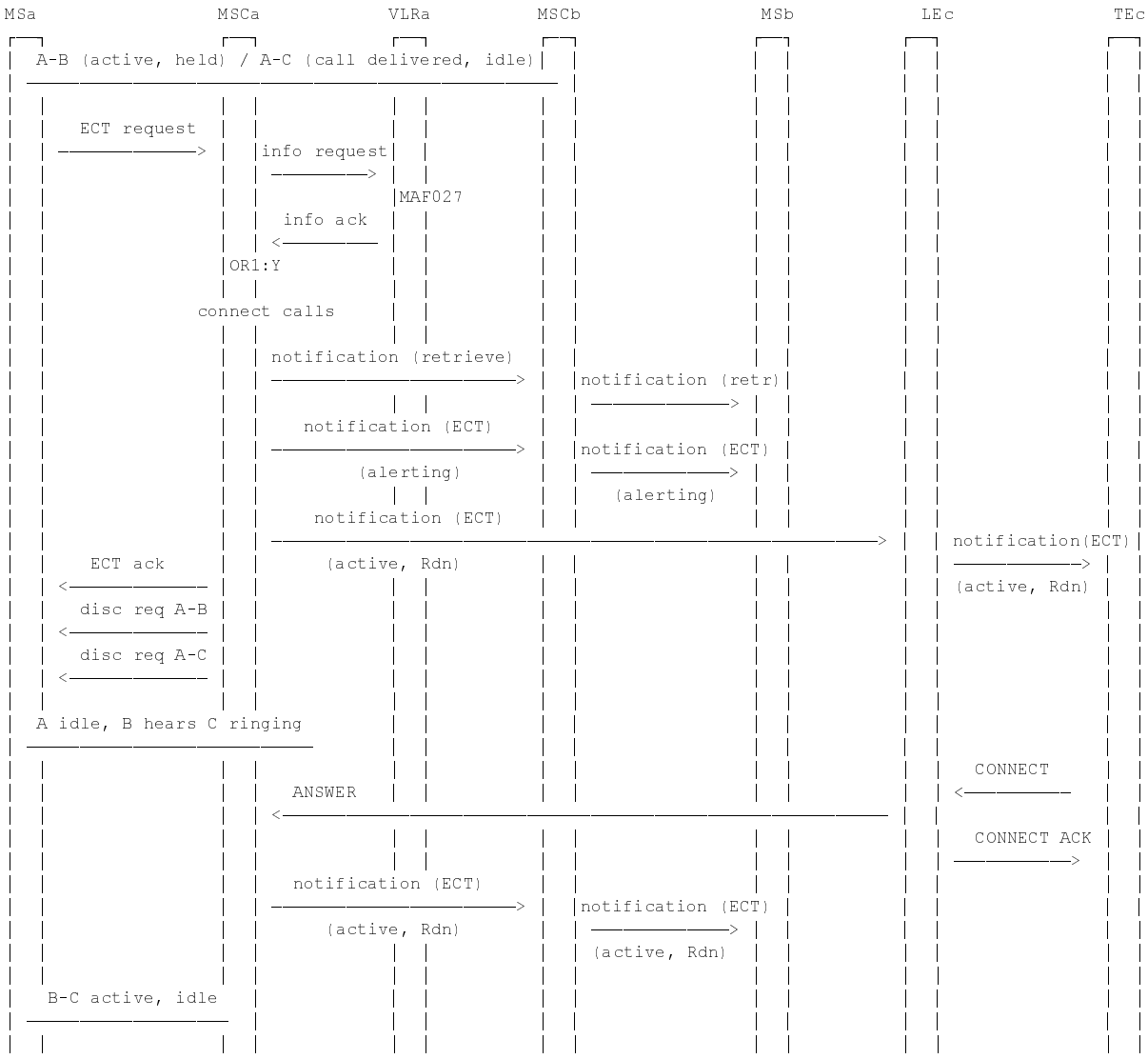


Figure 6: Overall SDL-diagram of explicit call transfer (one call answered, the other alerting) (page 2 of 2)



NOTE: OR1: Checks in VLR and MSC ok? (Y: yes N: no).

Figure 7: Information flow for failed explicit call transfer request (one call answered, the other alerting)



NOTE: OR1: Checks in VLR and MSC ok? (Y: yes N: no).

Figure 8: Information flow for successful explicit call transfer (one call answered, the other alerting)

4.3 Interaction with other supplementary services

4.3.1 Line Identification services

Tables 1 to 4 indicate the information to be provided in the Notification Indicator (NI) and the Redirection Number (Rdn) when the subscribers B and C are notified. Call states refer to the situation before ECT invocation. At that time one of the calls is on hold.

If user B was the called subscriber in the call A-B, table 1 applies to the information supplied to subscriber C. If user B was the calling subscriber in the call A-B, table 2 applies to the information supplied to subscriber C.

Mobile subscriber A has an active call to subscriber B and:

- puts the active call on hold and calls subscriber C, table 3 applies to the information supplied to subscriber B;
- receives and accepts a call from subscriber C (by putting B on Hold), table 4 applies to the information supplied to subscriber B.

Table 1: Mobile subscriber A was calling subscriber B, puts B on hold and calls subscriber C

Call states	COLR indication received from subscribers B's network	Information provided to C
A-B Active A-C Active / Alerting	Indicated "allowed"	At time of transfer: NI: "call transferred, active" Rdn: PI = allowed, LI of B
A-B Active A-C Active / Alerting	Indicated "restricted"	At time of transfer: NI: "call transferred, active" Rdn: PI = restricted (note 1)
A-B Active A-C Active / Alerting	No indication received (e.g. interworking)	At time of transfer: NI: "call transferred, active" Rdn: PI = not available

Table 2: Mobile subscriber A was called by subscriber B, puts B on hold and calls subscriber C

Call states	CLIR indication received from subscribers B's network	Information provided to C
A-B Active A-C Active / Alerting	Indicated "allowed"	At time of transfer: NI: "call transferred, active" Rdn: PI = allowed, LI of B
A-B Active A-C Active / Alerting	Indicated "restricted"	At time of transfer: NI: "call transferred, active" Rdn: PI = restricted (note 1)
A-B Active A-C Active / Alerting	No indication received (e.g. interworking)	At time of transfer: NI: "call transferred, active" Rdn: PI = not available

NOTE 1: If the subscriber C has CLIP Override Category then the following information is carried in the Redirection number: PI = restricted, LI of B.

Table 3: Mobile subscriber A puts the call to B on hold and calls subscriber C

Call states	COLR indication received from subscribers C's network	Information provided to B
A-B Active A-C Active	Indicated "allowed"	At time of transfer: NI: "call transferred, active" Rdn: PI = allowed, LI of C
A-B Active A-C Active'	Indicated "restricted"	At time of transfer: NI: "call transferred, active" Rdn: PI = restricted (note 2)
A-B Active A-C Active	No indication received (e.g. interworking)	At time of transfer: NI: "call transferred, active" Rdn: PI = not available
A-B Active A-C Alerting	Indicated "allowed" at receipt of CONNECT by subscriber C	At time of transfer: NI: "call transferred, alerting" At subscribers C CONNECT: NI: "call transferred, active" Rdn: PI = allowed, LI of C
A-B Active A-C Alerting	Indicated "restricted" at receipt of CONNECT by subscriber C	At time of transfer: NI: "call transferred, alerting" At subscribers C CONNECT: NI: "call transferred, active" Rdn: PI = restricted (note 2)
A-B Active A-C Alerting	No indication received at receipt of CONNECT by subscriber C (e.g. interworking)	At time of transfer: NI: "call transferred, alerting" At subscribers C CONNECT: NI: "call transferred, active" Rdn: PI = not available

Table 4: Mobile subscriber A was called by subscriber C and accepts the call by putting subscriber B on hold

Call states	CLIR indication received from subscriber C's network	Information provided to B
A-B Active A-C Active	Indicated "allowed"	At time of transfer: NI: "call transferred, active" Rdn: PI = allowed, LI of C
A-B Active A-C Active	Indicated "restricted"	At time of transfer: NI: "call transferred, active" Rdn: PI = restricted (note 2)
A-B Active A-C Active	No indication received (e.g. interworking)	At time of transfer: NI: "call transferred, active" Rdn: PI = not available

NOTE 2: If the subscriber B was called by subscriber A and has CLIP Override Category, or if subscriber B called subscriber A and has COLP Override Category then the following information is carried in the Redirection number: PI = restricted, LI of C

4.3.2 Call Forwarding Unconditional (CFU)

No impact.

4.3.3 Call Forwarding on mobile subscriber Busy (CFB)

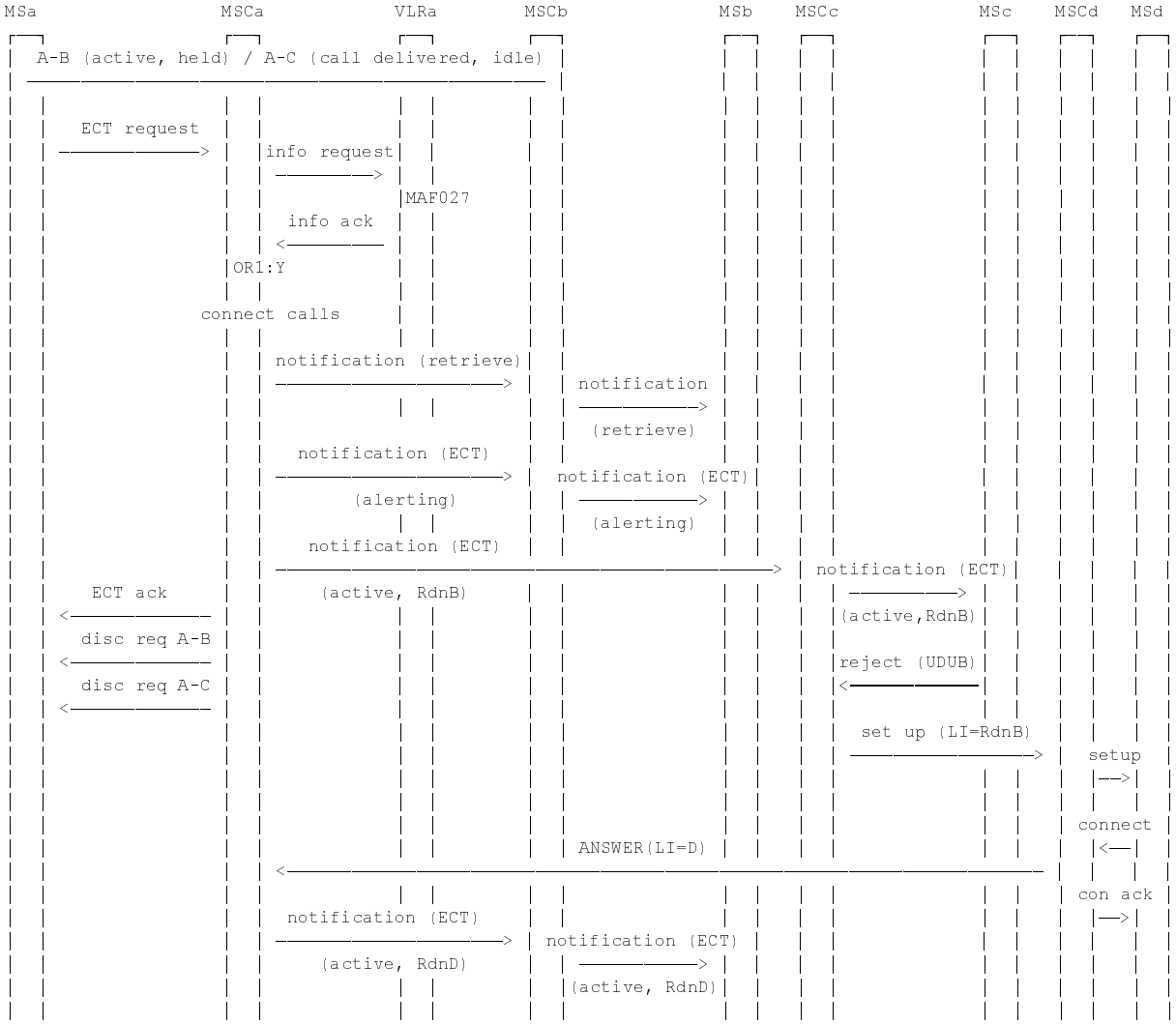
4.3.3.1 Call Forwarding on mobile subscriber Busy due to Network Determined User Busy (NDUB)

No impact.

4.3.3.2 Call Forwarding on mobile subscriber Busy due to User Determined User Busy (UDUB)

When subscriber A transfers the forwarded call there is no impact.

When subscriber C forwards the transferred call to the forwarded-to subscriber D due to UDUB the line identity information of the subscriber B that was received by the subscriber C in the ECT invocation notification shall be sent as calling line identity to the forwarded-to subscriber D instead of the line identity of the subscriber A. The corresponding information flow is given in figure 9. For the line identity information sent to the subscriber B after the call is answered by the forwarded-to subscriber D the table 5 applies.



NOTE: OR1: Checks in VLR and MSC ok? (Y: yes N: no)

Figure 9: Information flow for interaction of explicit call transfer (one call answered, the other alerting) with call forwarding

Table 5: Subscriber C forwards the transferred call to the subscriber D

Call states	COLR indication received from subscribers D's network	Information provided to B
A-B Active A-C Alerting, forwarded to D	Indicated "allowed" at receipt of CONNECT by subscriber D	At time of transfer: NI: "call transferred, alerting" At subscribers D CONNECT: NI: "call transferred, active" Rdn: PI = allowed, LI of D
A-B Active A-C Alerting, forwarded to D	Indicated "restricted" at receipt of CONNECT by subscriber D	At time of transfer: NI: "call transferred, alerting" At subscribers D CONNECT: NI: "call transferred, active" Rdn: PI = restricted (note 1)
A-B Active A-C Alerting, forwarded to D	No indication received at receipt of CONNECT by subscriber D (e.g. interworking)	At time of transfer: NI: "call transferred, alerting" At subscribers D CONNECT: NI: "call transferred, active" Rdn: PI = not available

NOTE 1: If the subscriber B was called by subscriber A and has CLIP Override Category, or if subscriber B called subscriber A and has COLP Override Category then the following information is carried in the Redirection number: PI = restricted, LI of D.

4.3.4 Call Forwarding on No Reply (CFNRy)

Same as the interaction between call forwarding on mobile subscriber busy due to UDUB and explicit call transfer as described in subclause 4.3.3.2.

Figure 9 applies except that call forwarding is invoked by the CFNRy timer expiry instead of reception of reject (UDUB) message.

For the line identity information sent to the subscriber B after the call is answered by the forwarded-to subscriber D the table 5 applies.

4.3.5 Call Forwarding on mobile subscriber Not Reachable (CFNRc)

No impact.

4.3.6 Call Waiting (CW)

No impact.

4.3.7 Call Hold (HOLD)

No impact.

4.3.8 Multi Party (MPTY)

The MSC/VLR shall reject any ECT request from the served subscriber of a MPTY call.

4.3.9 Closed User Group (CUG)

Closed user group restrictions shall be met between users when the first call is set up.

Similarly, closed user group restrictions shall also be met between users when setting up the second call.

Finally, for successful explicit call transfer the served mobile subscriber must use the same CUG-Interlock code for both calls. The same rule shall applied regardless of being two MO calls, two MT calls or one MO and one MT call.

4.3.10 Advice of Charge (AoC) services

No impact.

4.3.11 Call Barring services

No impact

4.3.12 Explicit Call Transfer (ECT)

It is required as a network option that the establishment of endless loops between subscriber A and subscriber B, both of them transferring the call to the other one, is prevented. The same loop prevention mechanism as in ISDN shall be used.

4.4 Information stored in the HLR

The following logical states are applicable for the Explicit Call Transfer service (refer to GSM 03.11 for an explanation of the notation):

Provisioning State	**Registration State**	**Activation State**	**HLR Induction State**
(Not Provisioned,	Not Applicable,	Not Active,	Not Induced)
(Provisioned,	Not Applicable,	Active and Operative,	Not Induced)

The HLR shall store the logical state of the Explicit Call Transfer service (which shall be one of the valid states listed above) on a per subscriber basis.

4.5 State transition model

Figure 10 shows the successful cases of transition between the applicable logical states of the Explicit Call Transfer service. The state changes are caused by actions of the service provider.

Note that error cases are not shown in the diagram as they normally do not cause a state change. Additionally, some successful requests may not cause a state change and are therefore not shown in the diagram.

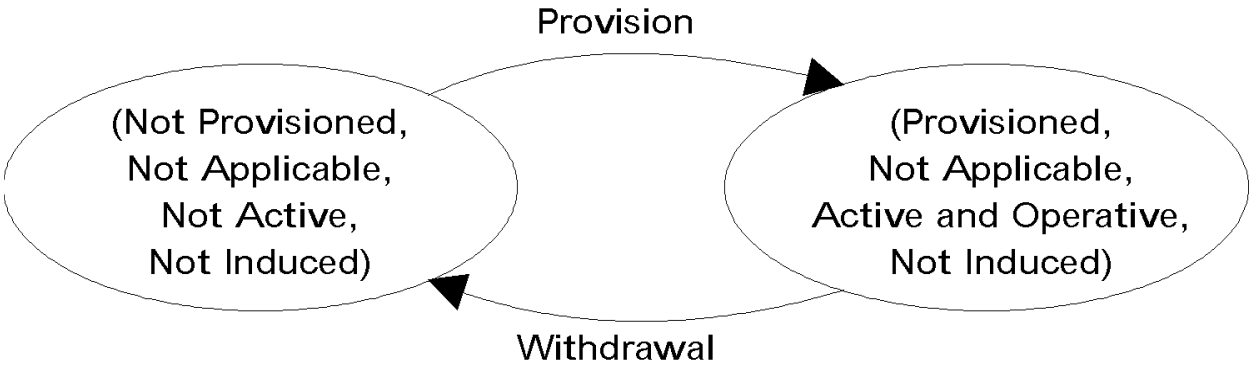


Figure 10: State transition model

4.6 Transfer of information from the HLR to the VLR

If the provisioning state for the Explicit Call Transfer service is "Provisioned" then when the subscriber registers on a VLR the HLR shall send that VLR information about the logical state of the Explicit Call Transfer service.

If the logical state of the Explicit Call Transfer service is changed while a subscriber is registered on a VLR then the HLR shall inform the VLR of the new logical state of the Explicit Call Transfer service.

4.7 Information stored in the VLR

For the supplementary service Explicit Call Transfer the VLR shall store the service state information received from the HLR.

4.8 Handover

Handover will have no impact on the control procedures and the operation of the service.

Annex A (informative): Change Request History

Status of Technical Specification GSM 03.91		
Date	**Version**	**Remarks**
		No Phase 1 version
June 1994	1.0.0	To SMG#11 for information
April 1995	4.0.0	TS approved by SMG#14
October 1995	5.0.0	TS transferred to Phase 2+
April 1996	5.0.1	CR 03.91-A001r1 (category D) (ECT) approved by SMG#18
June 1996	5.0.2	CR 03.91-A003 (category D) (ECT) approved by SMG#19
SMG#27	6.0.0	Release 1997 version

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